EX 9

Forward Chaining
Exercise 1: Simple Inference
:- dynamic known/1.
% Initial knowledge
start :- assert(known(a)), assert(known(b)), infer.

% Rule
infer :- known(a), known(b), \+ known(c), assert(known(c)), write('Inferred c'), nl.
% Query
% ?- start.
Output:
Inferred c

Exercise 2: Fire Alert System
:- dynamic known/1.
% Rule to infer fire
check :- known(smoke), known(heat), assert(known(fire)), write('Fire Alert!'), nl.
% Start rule
start :- assert(known(smoke)), assert(known(heat)), check.
% Query
% ?- start.
Output:
Fire Alert!

Exercise 3: Water Purity Detection
:- dynamic known/1.
% Rule to infer pollution
infer_pollution :- known(cloudy_water), known(bad_smell), assert(known(polluted)),
write('Water is polluted'), nl.
start :- assert(known(cloudy_water)), assert(known(bad_smell)), infer_pollution.
% Query
% ?- start.
Output:
Water is polluted